

Chapter - 2

Biological Classification

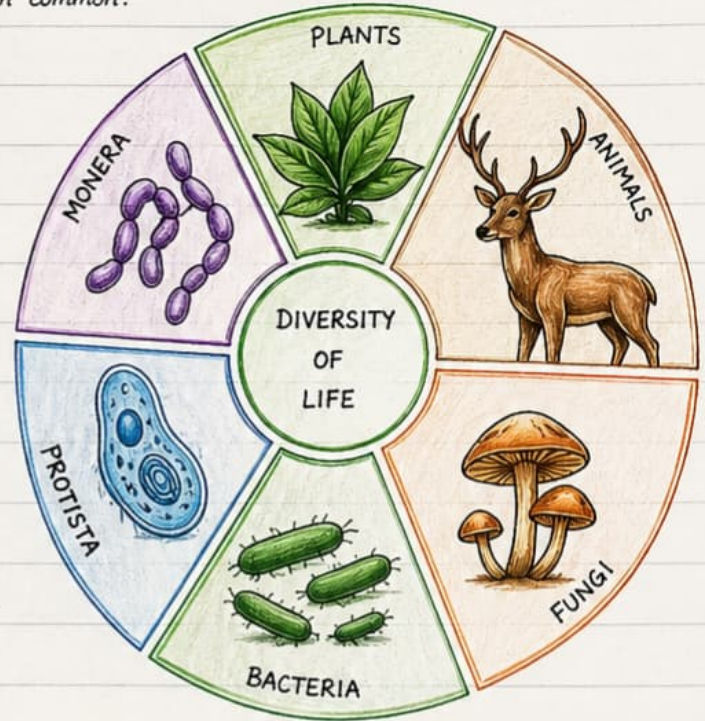
NEET UG

Class - 11th

"Diversity is the one true thing we all have in common."

1. WHAT IS BIOLOGICAL CLASSIFICATION ?

- Biological classification is the systematic arrangement of **living organisms** into groups and categories based on their **similarities and differences**.
- Helps in **easy study**, **identification** and understanding of vast diversity of life.
- It is a **hierarchical system** - organisms are placed in groups within groups.



2. NEED FOR CLASSIFICATION

- To make the study of organisms **easy**.
- To **identify** organisms accurately.
- To know **relationships** among organisms.
- To **predict characteristics** of organisms.
- To avoid **confusion** due to **local or common names**.



Over 18 million species of organisms exist on Earth. Classification helps us to organize this huge diversity in a meaningful way.

3. BASIS OF CLASSIFICATION

- Classification is based on **similarities and differences** in :

Morphology
(Structure)

Anatomy
(Internal structure)

Physiology
(Functions)

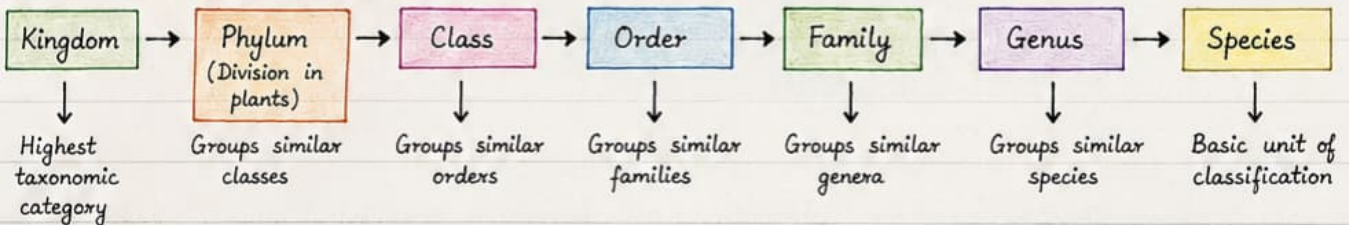
Biochemistry
(Chemical composition)

Genetics
(DNA, RNA)

Evolution
(Ancestral relationship)

4. TAXONOMIC CATEGORIES

(Hierarchy of classification) →



Example : Plantae → Angiospermae → Dicotyledonae → Rosales → Fabaceae → Pisum → Pisum sativum
(Kingdom) (Division) (Class) (Order) (Family) (Genus) (Species)

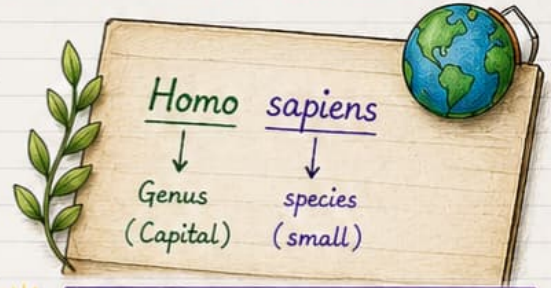
★ Species is the fundamental unit.
All organisms of a species can interbreed and produce fertile offspring.

Remember!
Similarities indicate relationship,
differences indicate diversity.

5. BINOMIAL NOMENCLATURE

- Each organism has a unique two-word scientific name.
- The first word is the Genus name and the second is the specific epithet (species name).
- Genus name starts with a capital letter.
- Species name starts with a small letter.
- Write in italics or underline separately.
- After first mention, the genus name can be abbreviated.

Given by → Carolus Linnaeus



Example: Mango → *Mangifera indica*
Human → *Homo sapiens*

6. TAXONOMIC AIDS

These are the tools or resources used by taxonomists for identification and classification of organisms.

(A) HERBARIUM

- A herbarium is a store house of dried and pressed plant specimens.
- Each specimen is properly labeled with details like name, place, date, habitat, etc.
- Example – National Herbarium (Howrah, India)



(B) MUSEUM

- A museum contains preserved specimens of animals, plants, microorganisms, fossils, etc.
- They help in studying the structure, diversity and evolution of organisms.
- Example – Indian Museum, Kolkata



(C) BOTANICAL GARDENS

- These are living collections of various plants grown for study and conservation.
- They help in the identification of plants.
- Example – Indian Botanic Garden, Howrah



(D) KEYS

- Keys are analytical tools used to identify organisms.
- They are based on a series of choices leading to the correct name.
- Two types –
 - Dichotomous key
 - Numerical key

Example of Dichotomous Key

- 1a. Leaves simple → go to 2
- 1b. Leaves compound → go to 3
- 2a. Leaves parallel → Plant A
- 2b. Leaves reticulate → Plant B

7. SOME IMPORTANT EXAMPLES OF CLASSIFICATION

Organism	Scientific Name	Genus	Family
Human	<i>Homo sapiens</i>	<i>Homo</i>	Hominidae
Mango	<i>Mangifera indica</i>	<i>Mangifera</i>	Anacardiaceae
Housefly	<i>Musca domestica</i>	<i>Musca</i>	Muscidae
Lion	<i>Panthera leo</i>	<i>Panthera</i>	Felidae
Rice	<i>Oryza sativa</i>	<i>Oryza</i>	Poaceae

★ Classification brings order in diversity and helps us to understand the living world better.

In short: Classification → Identification → Nomenclature → Understanding diversity

Learn Names, Know Nature, Save Future. ♥

8. FIVE KINGDOM CLASSIFICATION

R. H. Whittaker (1969) proposed a five kingdom classification based on cell structure, body organization, mode of nutrition and reproduction.

The Five Kingdoms

- Monera** → Prokaryotic, unicellular, e.g., Bacteria, Cyanobacteria
- Protista** → Eukaryotic, mostly unicellular, e.g., Amoeba, Paramecium, Euglena
- Fungi** → Eukaryotic, mostly multicellular, non-green, absorptive nutrition, e.g., Mushroom, Yeast, Penicillium
- Plantae** → Eukaryotic, multicellular, photosynthetic, cell wall present, e.g., Moss, Fern, Mango
- Animalia** → Eukaryotic, multicellular, no cell wall, ingestive nutrition, e.g., Hydra, Earthworm, Human

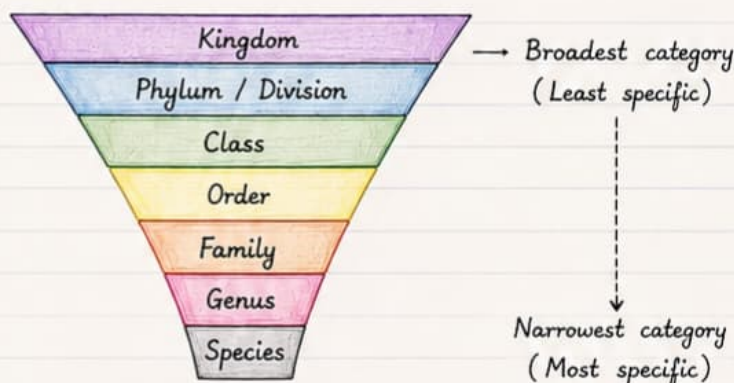
MONERA	PROTISTA	FUNGI	PLANTAE	ANIMALIA
				
• Prokaryotic	• Eukaryotic	• Eukaryotic	• Eukaryotic	• Eukaryotic
• Unicellular	• Mostly unicellular	• Mostly multicellular	• Multicellular	• Multicellular
• Cell wall (peptidoglycan)	• Some with cell wall	• No chlorophyll	• Cell wall (cellulose)	• No cell wall
• Autotrophic/Heterotrophic	• Autotrophic/Heterotrophic	• Cell wall (chitin)	• Chlorophyll present	• No chlorophyll
• Asexual reproduction	• Asexual & sexual reproduction	• Heterotrophic (absorptive)	• Autotrophic	• Ingestive nutrition
		• Spore formation	• Sexual & asexual reproduction	• Sexual reproduction

9. HIERARCHY OF TAXONOMY

Taxonomic categories are arranged in a hierarchical order from broad to specific.

★ Example (Human - Homo sapiens)

Kingdom	→	Animalia
Phylum	→	Chordata
Class	→	Mammalia
Order	→	Primates
Family	→	Hominidae
Genus	→	Homo
Species	→	Homo sapiens



10. IMPORTANCE OF BIOLOGICAL CLASSIFICATION

- It helps in organizing vast diversity of living organisms.
- It helps in easy identification of organisms.
- It shows relationships and evolutionary links.
- It helps in communication by using universally accepted names.
- It helps in conservation of biodiversity.

QUICK POINTS

- ★ Classification is dynamic.
- ★ New organisms may change place when new information is available.
- ★ It is a continuous process.



SHORTCUT TO REMEMBER (5 KINGDOMS)

				
M	P	F	P	A
Monera	Protista	Fungi	Plantae	Animalia

NEET TIP

Always remember the base of Five Kingdom Classification → Cell type, Body organization, Mode of nutrition and Reproduction.



The aim of classification is not to put organisms into boxes, but to show the beautiful connections of life.



11. KEY CONCEPTS – QUICK REVISION

- ★ Classification is a **systematic** process.
- ★ It is based on **similarities** and **differences**.
- ★ **Taxonomy** provides names to organisms to avoid confusion.
- ★ **Binomial nomenclature** gives each species a unique name.
- ★ The **five kingdom** system is widely accepted and useful.
- ★ **Hierarchy** in classification helps in proper arrangement from broad to specific.
- ★ Classification helps us understand **evolution** and conserve **biodiversity**.

12. EXAMPLE OF CLASSIFICATION – MANGO (*Mangifera indica*)

Kingdom → Plantae
 Division → Angiospermae
 Class → Dicotyledonae
 Order → Sapindales
 Family → Anacardiaceae
 Genus → *Mangifera*
 Species → *indica*

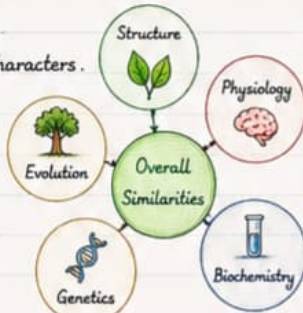


★ The scientific name of mango is *Mangifera indica*.

13. NATURAL VS ARTIFICIAL CLASSIFICATION

Natural Classification

- Based on a large number of characters.
- Considers overall similarities.
- Shows natural relationships.
- More reliable and stable.
- Example: Five kingdom system.



Artificial Classification

- Based on few characters.
- Considers superficial similarities.
- Does not show natural relationships.
- Less reliable.
- Example: Based only on habitat or size.



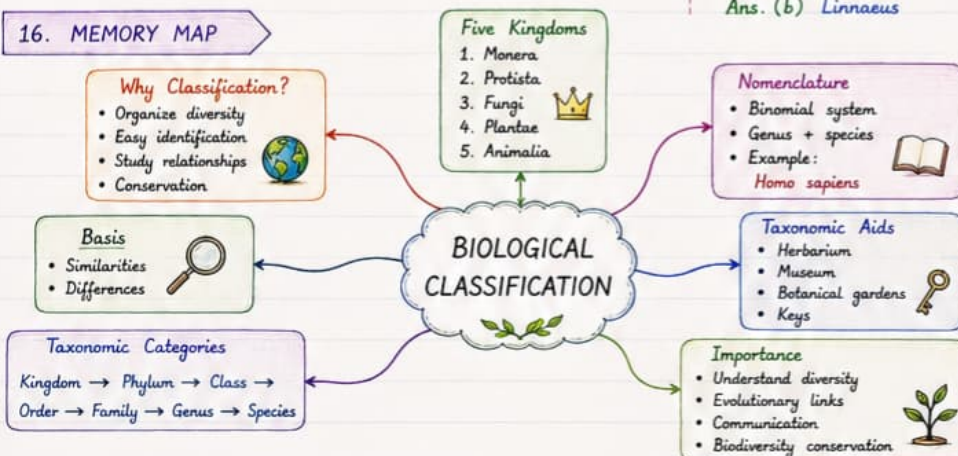
14. SOME IMPORTANT TERMS

- **Taxonomy**: Science of identification, naming and classification.
- **Systematics**: Study of diversity of organisms and their evolutionary relationships.
- **Nomenclature**: System of giving names.
- **Identification**: Recognizing an organism and assigning it a proper name.
- **Biodiversity**: Variety and variability among living organisms.

15. PYQ CORNER (NEET UG)

- Q1. Which of the following is not a taxonomic category?
 (a) Genus (b) Family (c) Species (d) Class
 Ans. (c) Species (Basic unit, not a category)
- Q2. Which one of the following is the correct sequence of taxonomic categories?
 (a) Species → Genus → Family → Order
 (b) Kingdom → Phylum → Class → Order
 (c) Class → Order → Family → Genus
 (d) Family → Order → Class → Genus
 Ans. (b) Kingdom → Phylum → Class → Order ...
- Q3. The father of taxonomy is
 (a) Aristotle (b) Linnaeus (c) Darwin (d) Haeckel
 Ans. (b) Linnaeus

16. MEMORY MAP



Remember! ★
 Names may change, but classification helps us know the true nature of life on Earth. ♥